

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

Critical Path Analysis

Section I – Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

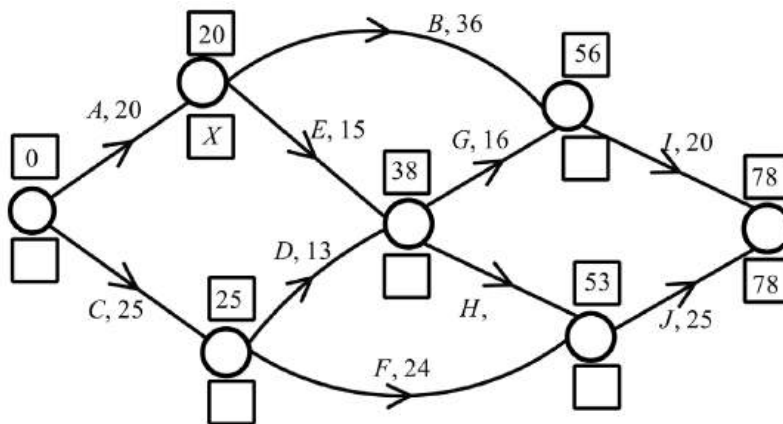
Section I

15 marks

Questions 1 – 3 refer to the following:

The diagram shows the activities required to complete a factory process.

A forward scan has been completed for the process.



1. The duration of activity *H* is missing. What value should appear?
(A) 13
(B) 14
(C) 15
(D) 16
2. When a backward scan is completed, what is the value of *X*?
(A) 20
(B) 22
(C) 24
(D) 25
3. What is the critical path?
(A) *ABJ*
(B) *AEIJ*
(C) *CDGJ*
(D) *CDHJ*

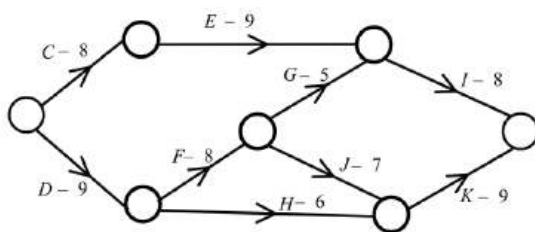
Questions 4 – 6 refer to the following:

The table below shows the activities and their prerequisites which make up a process.

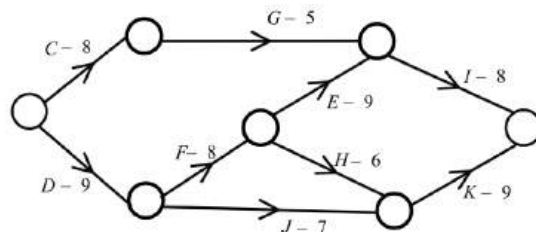
Activity	Immediate prerequisite(s)	Duration in days
Start	-	-
C	-	8
D	-	9
E	C	9
F	D	8
G	F	5
H	D	6
I	E, G	8
J	F	7
K	J, H	9
End	I, K	-

4. Which diagram could represent this table?

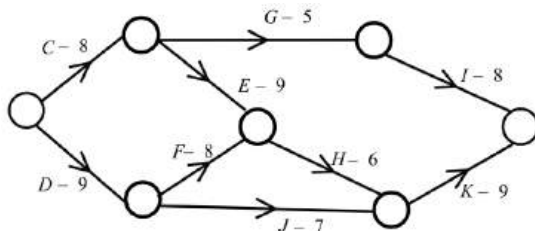
(A)



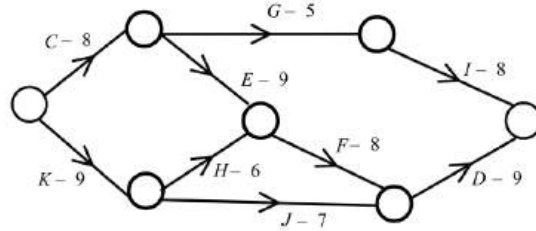
(B)



(C)



(D)



5. What is the earliest start time for activity I?

(A) 17

(B) 22

(C) 24

(D) 25

6. What is the critical time for the entire process?

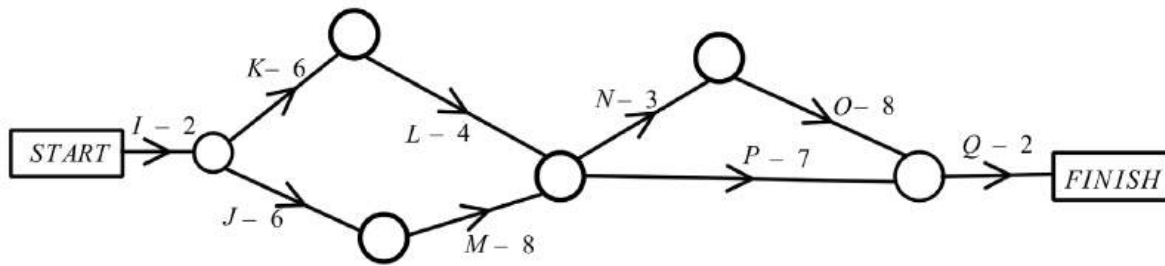
(A) 33

(B) 35

(C) 38

(D) 40

Questions 7 – 9 refer to the diagram below.



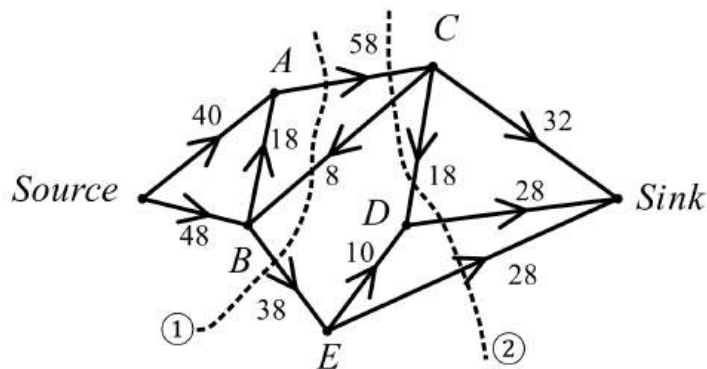
The diagram shows the activities required to progress from the start to the finish of a process and each activity has its duration (in days) shown beside it.

7. What is the earliest starting time (*EST*) for activity *Q*?
 - (A) 23
 - (B) 27
 - (C) 28
 - (D) 29
8. Which is the critical path?
 - (A) *IKLPQ*
 - (B) *IJMPQ*
 - (C) *IJMNOQ*
 - (D) *IKLNOQ*
9. What is the float time for activity *K*?
 - (A) 1 day
 - (B) 2 days
 - (C) 3 days
 - (D) 4 days

Questions 10 – 12 refer to the following.

A network flow diagram is shown below.

Two cuts, labelled ① and ②, have been drawn on the diagram.

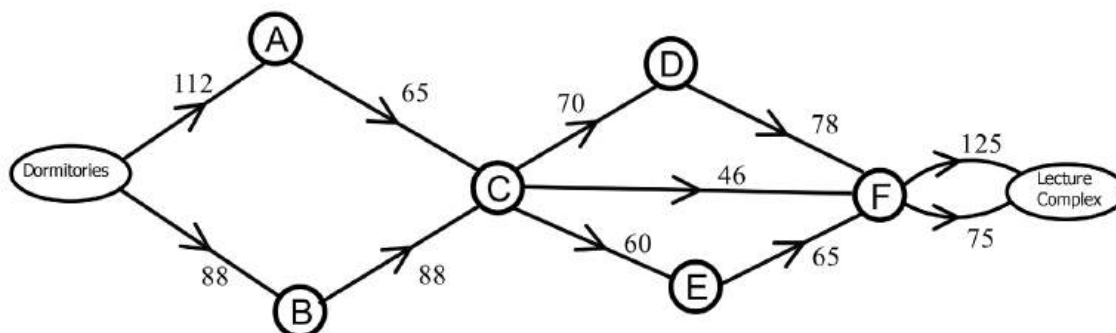


10. What is the total weight of the cut labelled ① ?
 - (A) 88
 - (B) 96
 - (C) 98
 - (D) 104
11. What is the total weight of the cut labelled ② ?
 - (A) 88
 - (B) 96
 - (C) 114
 - (D) 140
12. What is the maximum flow of this network?
 - (A) 88
 - (B) 96
 - (C) 104
 - (D) 114

Questions 13 – 15 refer to the following.

The network diagram represents a system of one-way paths allowing students to walk from the dormitories to the lecture complex of a university, passing through a series of intersections, labelled A to F.

The number on the edge of each path indicates the number of students that can travel on it per hour.



13. What is the excess capacity of the path from the dormitories to intersection A?
(A) 47
(B) 65
(C) 112
(D) 177
14. What is the maximum flow from the dormitories to the lecture complex?
(A) 125
(B) 153
(C) 173
(D) 176
15. Which extra path(s) would increase the network flow to its maximum?
(A) A path from the dormitories to C with capacity 47
(B) A path from C to F with capacity 24
(C) Paths from the dormitories to C with capacity 24 and from C to F with capacity 47
(D) Paths from the dormitories to C with capacity 47 and from C to F with capacity 24

Western Mathematics

Mathematics Standard 2

Critical Path Analysis

TOPIC REVISION QUESTIONS - Section II

Instructions

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

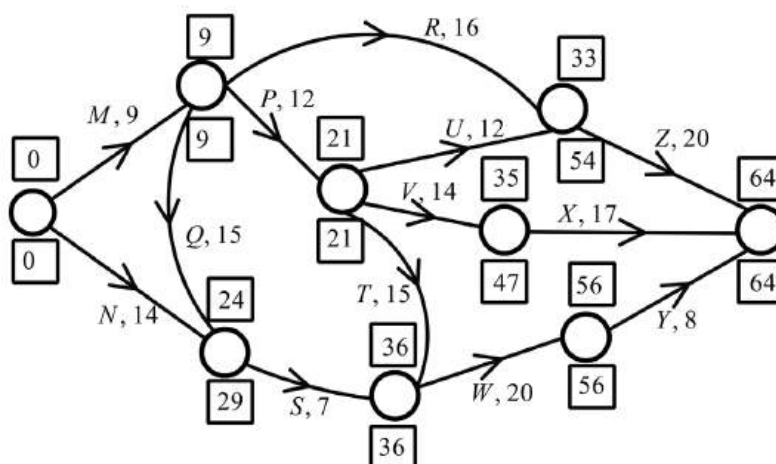
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 16 (5 marks)

Marks

The graph below shows the activities involved in a project with the times (in weeks) for each activity.

Forward and backward scans have been completed on the graph.



- List the immediate prerequisite activities of activity Z. 1
.....
- List the activities in the critical path. 1
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- What is the time of the critical path? 1
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- What is the float of activity X? 1
.....
- How many days delay can occur on activity N before it impacts the critical time? 1
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Marks

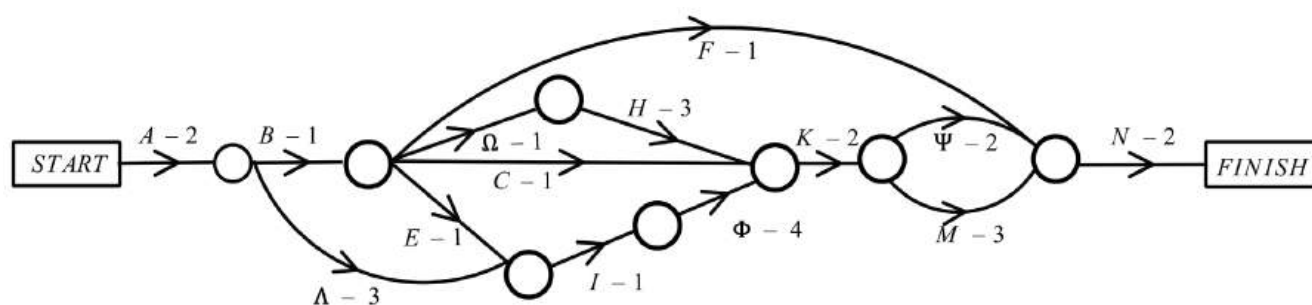
Question 17 (6 marks)

The table below outlines the activities involved in constructing a wooden deck.

Activity	Description	Duration	Prerequisites
A	Draw Plans	2	-
B	Calculate Materials Needed	1	A
C	Buy Hardware	1	B
D	Order Timber	1	B
E	Order Concrete	1	B
F	Buy Paint and Stain	1	B
G	Dig Foundations	3	A
H	Receive Timber Delivery	3	D
I	Pour Concrete in Foundations	1	E G
J	Wait for Concrete to Cure	4	I
K	Attach Timber Bearers and Joists	2	C J H
L	Lay Decking Boards	2	K
M	Build Railings and Steps	3	K
N	Stain and Paint Deck and Railings	2	F L M

- (a) A critical path analysis diagram has been drawn for the activities above.
Write the letters from the table that should replace Λ , Ω , Φ and Ψ .

2



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Marks

- (b) Complete a forward and backward scan on the diagram. (Write on the diagram.)
Hence determine the critical path and the critical time for the project.

3

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- (c) Assuming all other activities run to schedule, and there is a delay in the timber delivery, how many days late could it be before it delays the project?

1

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Question 18 (5 marks)

A project requires activities A to G to be completed, as shown in the table.

<i>Activity</i>	<i>Immediate prerequisite(s)</i>	<i>Duration in days</i>
A	–	5
B	–	24
C	B	?
D	A	?
E	D	20
F	E, G	15
G	A	40
H	C, F	?

The critical time for the project is 65 days and the critical path includes activities A , G , F and H . The float for E is 2 days and the float for C is 10 days.

- (a) Draw a network diagram to represent the table.

2

- (b) Find a possible duration for each of the activities C , F and H .

3

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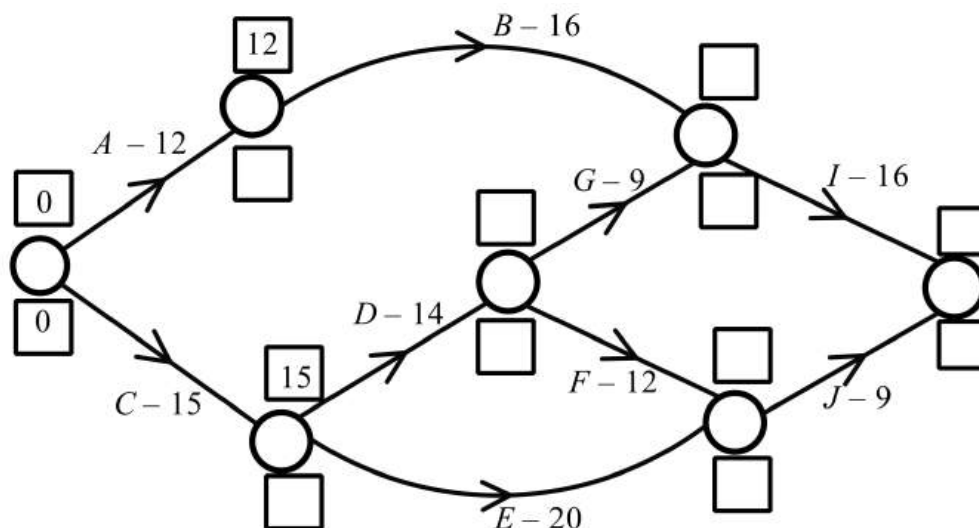
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Question 19 (5 marks)

This graph shows the time (in days) needed for each activity involved a project.

A forward scan has been started as part of the process to find the critical path for the diagram below.



- (a) Complete the forward scan and a backward scan on the network.
(Write your values in the spaces on the diagram.) 2

- (b) List the activities in the critical path and the time for this. 2

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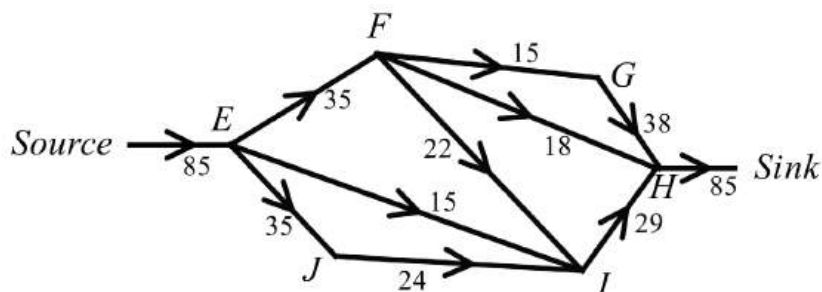
- (c) If activity *A* runs over by 4 days, how many days could activity *B* run over without increasing the time of the critical path? 1

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Question 20 (6 marks)

The diagram shows the network of pipes that convey gas from a source via six nodes to a sink.

Each pipes capacity in Gigalitres/hour is shown.



- (a) Determine the maximum flow through the network.

3

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- (b) The gas company has enough resources to upgrade two of the pipes in the network to a higher capacity.

3

Name two pipes that could be upgraded and their new capacity, so that the full capacity of the source can pass through the network.

Outline the effect that your changes would have on the flow in the network.

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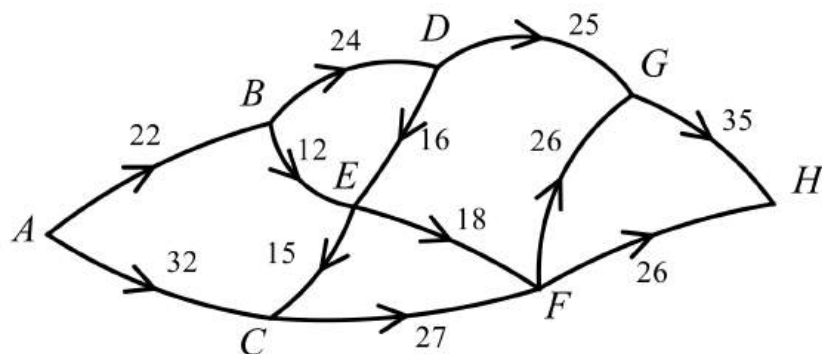
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Question 21 (5 marks)

An apparatus has a network of ducts which carry lubricant to several component parts.

The source of the lubricant is at A and a sink is at H .

The diagram shows the capacity of each duct with its capacity (in ml per minute).



- (a) How much lubricant could flow to component C in any given minute?

1

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- (b) Determine the maximum flow for the network of ducts.

2

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- (c) Component A can produce 60 ml/h of lubricant.

2

What single extra duct (which does not duplicate an existing duct) could be added to ensure that all the lubricant produced is able to flow through the network?

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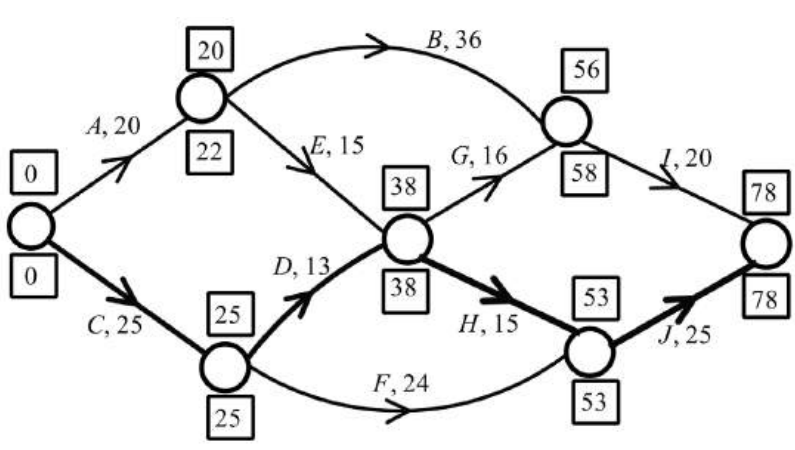
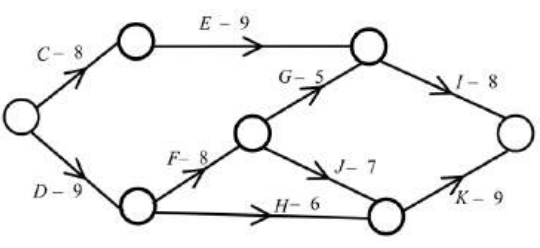
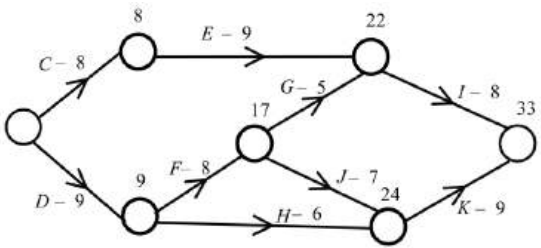
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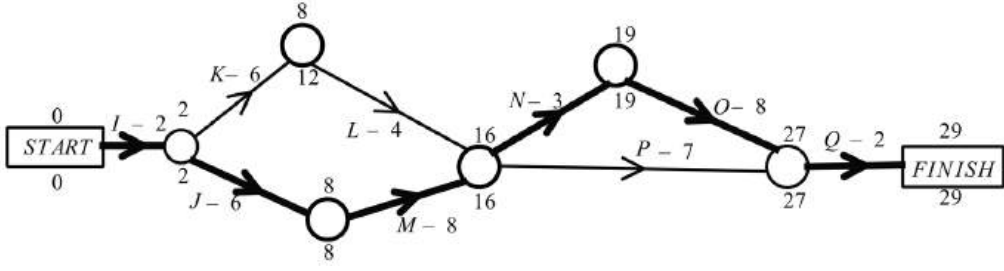
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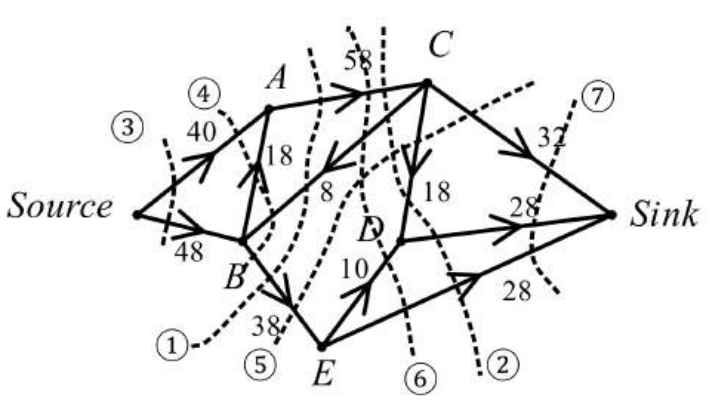
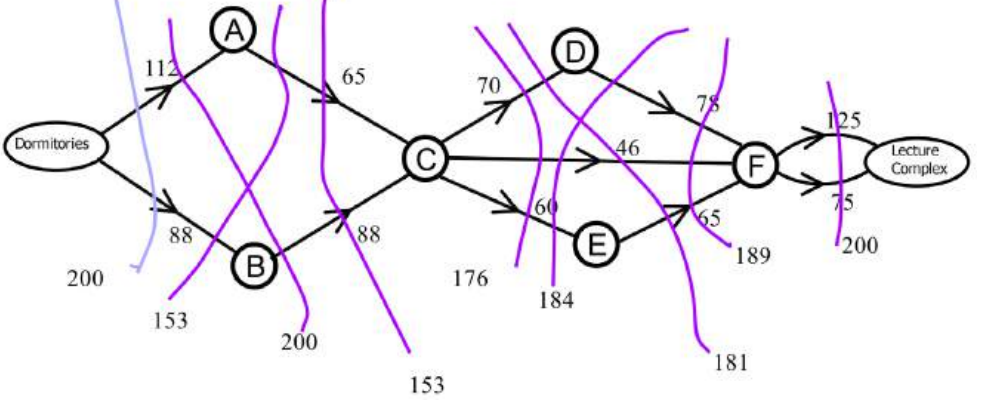
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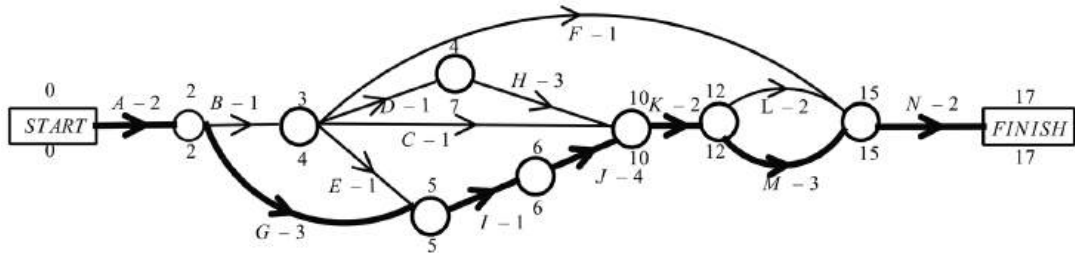
End of Revision

Western Mathematics
Critical Path Analysis
Revision
Section I - Worked Solutions

No	Working	Answer
1	As <i>H</i> lies between points whose forward scan are 53 and 38 $H = 53 - 38 = 15$	C
2	Completing the backward scan below, $X = 22$	B
3	The critical path passes through the points where the forward and backward scans give the same value. So <i>CDHJ</i> . 	D
4	The diagram below matches the table. 	A
5	EST for <i>I</i> is 22 	B

6	The critical time is 33 days, the value at the end of the last 2 activities.	A
7	 The EST is above the vertex preceding the activity, 27 days.	B
8	The critical path is highlighted, <i>IJMNOQ</i> .	C
9	The difference between the EST and LST for K is $12 - 8 = 4$ days	D
10	The cut labelled ① has weight equal to all the paths going from the source side to the sink side that cross the cut. Weight = $38 + 58 = 96$ CB goes from sink side to source side.	B
11	The cut labelled ② has weight equal to all the paths going from the source side to the sink side that cross the cut. Weight = $58 + 28 + 28 = 114$ CB and CD go from sink side to source side.	C

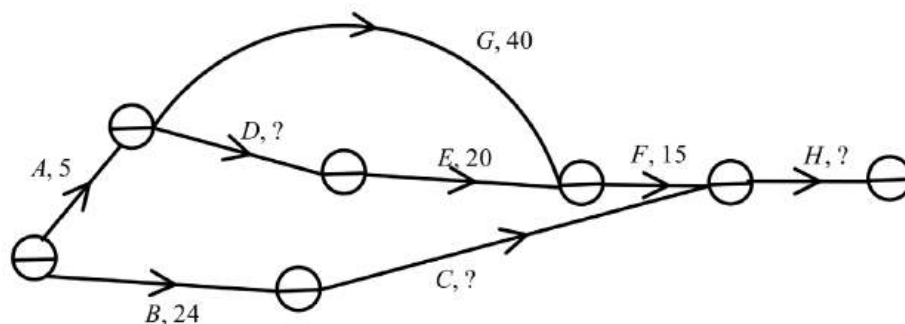
12	Other cuts are drawn and totalled.  Weight ③ = $48 + 40 = 88$ Weight ④ = $38 + 18 + 40 = 96$ Weight ⑤ = $38 + 18 + 32 = 88$ Weight ⑥ = $28 + 10 + 58 = 96$ Weight ⑦ = $28 + 28 + 32 = 88$ Maximum flow = minimum cut = 88	A
13	The path from the dormitories to intersection A has a capacity of 112 students per hour, but only 65 students per hour can proceed from A to C, so in fact only 65 can go from Dorm to A. So the path has an excess of $112 - 65 = 47$	A
14	 The minimum cut is 153, so this is the maximum flow.	B
15	The maximum flow in any cut is 200. The flow between the dorms and C could be increased from 153 to 200 by adding a path from the dormitories to C with capacity 47 The flow between C and F is limited to 176 this could be increased to 200 with a path from C to F with capacity 24	D

Critical Path Analysis Section II Solutions		Marks
Question 16		
(a)	R and U	1
(b)	MPTWY	1
(c)	64 weeks	1
(d)	$\text{Float} = \text{LST} - \text{EST} = 47 - 35 = 12$	1
(e)	N is a prerequisite for S. S cannot start till day 24 and can start as late as day 29 without impacting the critical path time. N is only scheduled to take 14 days, and could take up to 29 days, meaning it could have 15 days delay.	1
Question 17		
(a)	$\Lambda = G$ as activity <i>G</i> is missing lasts 3 days $\Omega = D$ as activity <i>D</i> is missing and lasts 1 day $\Phi = J$ as activity <i>J</i> is missing lasts 4 days $\Psi = L$ as activity <i>L</i> is missing and lasts 2 days	2 marks
(b)	 The critical path is A G I J K M N with a time of 17 days.	3 marks
(c)	H is receiving the timber delivery and has a float of $7 - 4 = 3$ days. This means that the timber could be up to 3 days late without effecting the overall progress, provided that B and D run to time as H depends on these.	1 mark

Question 18

(a)

2 marks



(b)

3 Marks

The sum of the critical path activities = $A + G + F + H = 65$

$$\text{So } 5 + 40 + 15 + H = 65$$

$$60 + H = 65$$

$$H = 5$$

Forward scanning, the earliest start time (EST) of E is $5 + D$ and backward scanning, the latest start time (LST) of E is $65 - 5 - 15 - 20 = 25$

Since the float for $E = 2$,

$$LST - EST = 2$$

$$25 - (5 + D) = 2$$

$$20 - D = 2$$

$$D = 20 - 2$$

$$D = 18$$

Forward scanning, the earliest start time (EST) of C is 24 and backward scanning, the latest start time (LST) of C is $65 - 5 - C = 60 - C$

Since the float for $C = 10$,

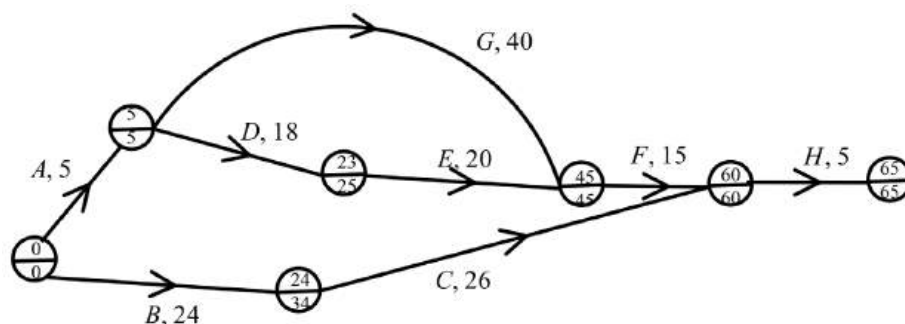
$$LST - EST = 10$$

$$60 - C - 24 = 10$$

$$36 - C = 10$$

$$C = 36 - 10$$

$$C = 26$$



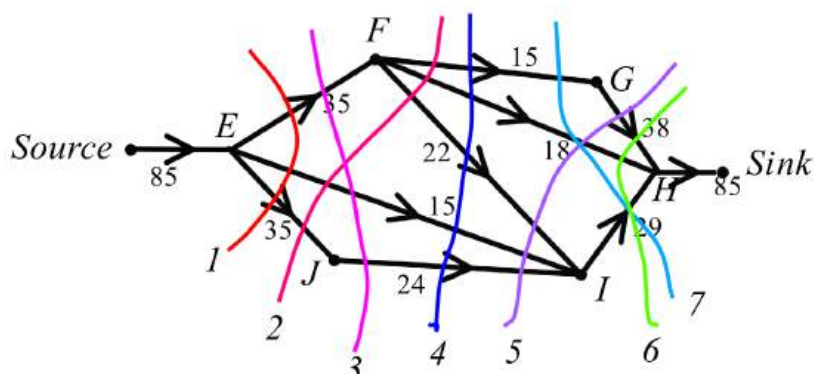
Question 19

(a)		3 marks
(b)	The Critical path is <i>CDGI</i> with a time of $15 + 14 + 9 + 16 = 54$	2 marks
(c)	The total of A and B is 28 days. They join the critical path after B. The float of B is $22 - 12 = 10$ If A runs over by 4 days, this increases the EST of B to 16, so float becomes $22 - 16 = 6$ days. So B count go at most 4 days over schedule	1 mark

Question 20

- (a) Seven cuts are drawn between the source and the sink.

3 marks



$$\text{Cut 1 flow} = 35 + 15 + 35 = 85$$

$$\text{Cut 2 flow} = 35 + 15 + 22 + 18 + 15 = 105$$

$$\text{Cut 3 flow} = 24 + 15 + 35 = 74$$

$$\text{Cut 4 flow} = 24 + 15 + 22 + 18 + 15 = 94$$

$$\text{Cut 5 flow} = 24 + 15 + 22 + 18 + 38 = 117$$

$$\text{Cut 6 flow} = 29 + 18 + 38 = 85$$

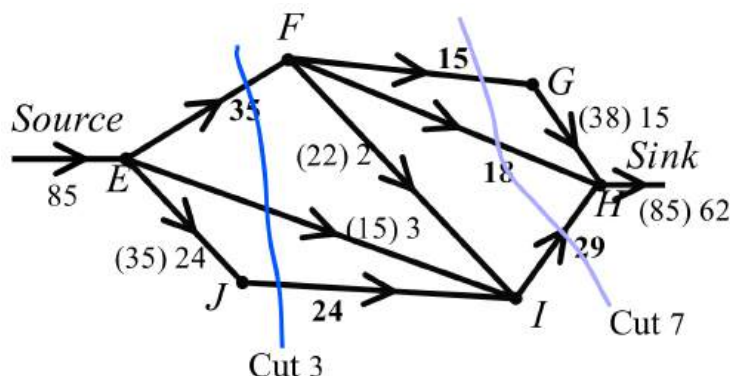
$$\text{Cut 7 flow} = 29 + 18 + 15 = 62$$

The maximum flow is the minimum cut, so 62 Gl per hr.

- (b) The pipes which are saturated are noted as **bold** and those whose flow is reduced by the inflow or outflow are changed from the bracketed value to the actual value that can flow through the pipe.

For example EJ has capacity 35 but can only carry 24 as it feeds into a single pipe (JI) with capacity 24.

FI has capacity 22 but as 35 flows from F into 3 pipes that have a total capacity of $15 + 18 + 22 = 55$, so after $FG + FH = 15 + 18 = 33$ there is only 2 left to go into FI.



The pipes which are saturated are those where an increase is possible that will benefit the whole network. The two cuts shown (3 and 7) are the two below 85 so they are the most likely to provide the increase needed.

Firstly on Cut 3, increasing JI to 35 allows all of the flow EJ to continue to I.

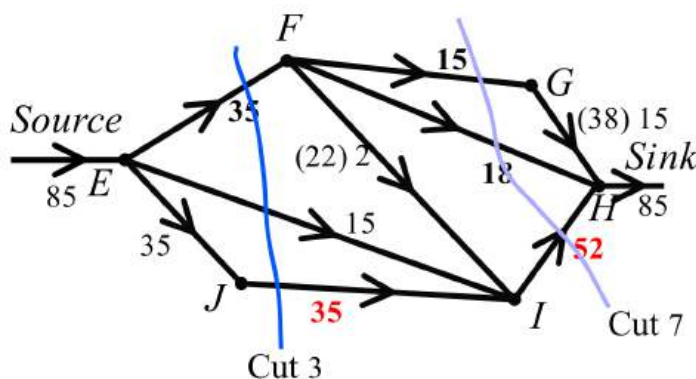
On Cut 7 the 3 pipes coming into I are limited by the 29 capacity pipe IH.

Pipes FI and EI are under capacity by $22 - 2 + 15 - 3 = 32$

Cut 7 is under the capacity of the source/sink by $85 - 62 = 23$ so increasing IH by 23 will allow this capacity to flow through. SO increase IH to $29 + 23 = 52$

Solution : Increase JI to 35 and increase IH to 52.

This allows EI and EJ to become saturated and hence the H to the sink to be saturated. FI and GH are still below capacity but this doesn't matter as the full capacity of the source and sink are being used.



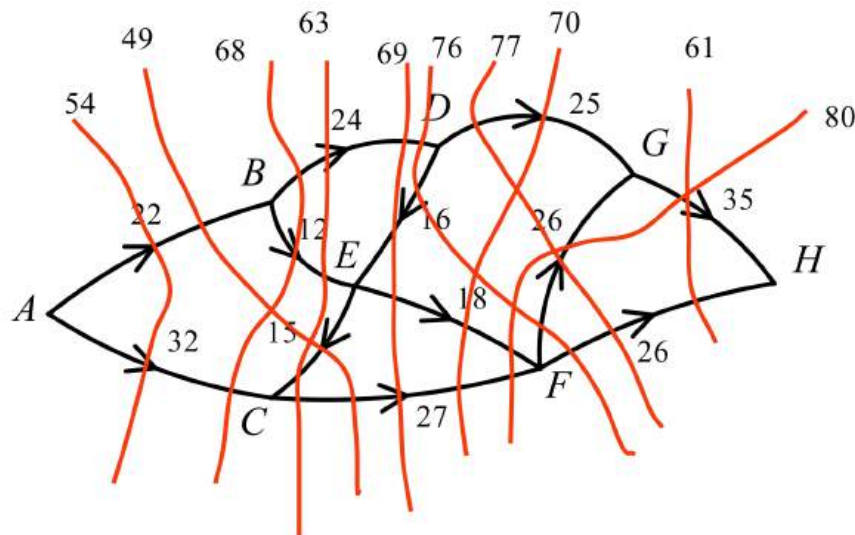
Question 21

- (a) C has ducts with capacity 32 and 15 flowing into it, but only 27 out, so the maximum that can enter is equal to the amount that can leave **27 ml**.

1 mark

(b)

2 marks



The minimum cut is 49 so the maximum flow is 49 ml/min

- (c) At the moment only 49 of the 60 ml can get through but both of the cuts of 49 and 54 are below 60, so we want to eliminate these with one extra duct. The 54 comes from A so we need to have extra from A and to go through the cut of 49.

2 marks

Going from A to B or A to C duplicates ducts, so going **A to E would be new and would need to carry at least 11 ml/min** to bring the flow to 60.

Going A to D would also achieve this as would A to F.